

Context Free Grammars

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LINGUIST 101 People and Their Language
Sections AA & AB
Fall 2026

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Plan for Today

- 1 Introductions (again)
- 2 Housekeeping
- 3 Context Free Grammars

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2 Housekeeping

3 Context Free Grammars

Introductions

- Name and pronouns (if you'd like to share)
- What are your weekend plans?

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Housekeeping #1

- Office Hours
 - TTh 1-2 & by appointment (gbcorrea@umass.edu)
 - N431A (Linguistics Department, ILC 4th floor)
- Section Website
 - <https://gabriel-correa.com/101-f26/>

Housekeeping #2

- Assignment #1 (Linguistic Diversity Survey) and Assignment #2 (CFGs) are due **next Monday (Sep 21)!**
- Make sure you submit them on time!
 - Late assignments will be worth 80% of the points
 - No assignments will be accepted after Wednesday, Sep 23
 - Email us if you need an extension!
- **Any questions** about Assignments #1 and #2?

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Infinity and recursion

- How do we know that language is **infinite**?
- Let's assume that the number of sentences that a language can have is **finite**!
 - If this is true, then we should be able to find the **longest sentence** in a language
 - Can you form a sentence longer than this one?
 - *John's mother's mother's friend's daughter is tall.*

Infinity and recursion

- **NP: Noun Phrase**
- $\text{NP} \rightarrow \text{NP}'\text{s NP}$
 - John's mother's mother's friend's daughter is tall.
 - John's mother's mother's friend is tall.
 - John's mother's mother is tall.
 - John's mother is tall.
 - John is tall.
- **Recursion:** defining something in terms of itself.

Infinity and recursion

- We say that the set of all sentences in a language is infinite
- But are sentences themselves infinite?



*Kyle thinks that Kriti believes
that Camila imagines that
Liam is certain that Alejandro
believes that Michi suspects...*

Context Free Grammars (CFGs)

- A context free grammar G has **four** components:

- 1 Start symbol: upper-case S
- 2 Non-terminal symbols: upper-case letters ($A, B, \dots$)
- 3 Terminal symbols: lower-case letters ($a, b, \dots$)
- 4 Rules

- 1 $S \rightarrow B A$

- 2 $A \rightarrow a$

- 3 $B \rightarrow B b \mid b$

$\mid$ = "or"

- Can this grammar generate bba ?
- Can this grammar generate aab ?

CFG Practice

- Let's try to do Grammar 1 and Grammar 7 in groups!

Have a great weekend!